GENETICS REVISION ANSWERS

1

(a) **He:** *Bb, Nn, Ff, Ss*

Ho: *SS, NN, ss, nn, ff, bb, BB, FF*

(b)

(i) *SS* — straight; *Ss* — straight; *ss* — curly

(ii) *FF* — freckles; *Ff* — freckles*; ff* — no freckles

(c)

(i) No glasses — *Nn* or *NN*; Needs glasses — *nn*

(ii) Blonde — *Bb* or *BB*; Not blonde — *bb*

(d)

(i)

		Tommy
	S	*s*
s	*Ss*	*ss*
s	*Ss*	*ss*

Susie

(ii) Genotypes — *Ss* or *ss*; Phenotypes — straight or curly
(iii) 50% chance of a child with straight hair
(iv) 50% chance of a child with curly hair

(e)

(i)

		Chuckie	
		n	*n*
Angelica	*N*	*Nn*	*Nn*
	n	*nn*	*nn*

(ii) Genotypes — *Nn* or *nn*; Phenotypes — no glasses or needs glasses
(iii) 50% chance of a child who needs glasses
(iv) 50% chance of a child who does not need glasses

(f) Didi's blood type is A and her genotype is *Ai*. She must have one allele for *A* in order to pass it on to Tommy and one recessive allele (*i*) that Dil would inherit in order for him to be O blood type (genotype *ii*).

2.

(a) Scott has the disease because he has inherited the X chromosome carrying the faulty allele from his mother who has the X-linked dominant condition. The fact that Scott's brother does not have the disease means that Scott's mother is heterozygous for the condition.

(b)

		Scott	
		X^W	Y
Female partner	X^w	X^WX^w	X^wY
	X^w	X^WX^w	X^wY

Scott's sons would not have the disease as they would inherit a normal copy of the X chromosome from their mother. Scott's daughters would be affected because they would inherit the faulty copy of the X chromosome from their father.

3.

	Mitosis	Meiosis
No. of cell divisions	1	2
No. of DNA replications	1	1
No. of daughter cells produced	2	4
No. of chromosomes in parent cell	46	46
No. of chromosomes in daughter cells	46	23
For growth and development? (Yes/No)	Yes	No
For reproduction? (Yes/No)	No	Yes

4.

1. The .....chromosomes..... of cells contains genes made of DNA.
2. Genes determine many .....physical..... characteristics (e.g. eye colour).
3. Diploid human cells contain 23 chromosomal pairs. ....gametes..... cells, however, contain only 23 chromosomes.
4. In a DNA molecule the amounts of thymine and adenine are always the same, and the amount of cytosine and .....guanine..... are always the same.
5. The Watson–Crick DNA model is a double .....helix..... like a twisted ladder.
6. Each step of the DNA helix is made up of two .....nitrogen..... bases and the bases are attached to the sugar units that are part of the sides of the ladder.
7. The steps of the DNA helix are composed of paired nitrogen bases. There is always an adenine base .....opposite..... a thymine base and a cytosine base opposite a guanine base.
8. Polypeptides are made up of units called .....amino..... acids. A set of three DNA bases codes for a particular amino acid.
9. During cell division (mitosis) the DNA needs to .....replicate..... to ensure the new cells have identical copies of the DNA.
10. In most instances the DNA is copied exactly during mitosis. In some cases .....errors..... occur and mutations result.
11. Gametes are formed by a type of cell division called .....meiosis.....
12. The fertilised egg has a .....diploid..... number of chromosomes.
13. X and Y are the sex chromosomes. Females have .....two..... X chromosomes and males have an X and a Y chromosome.
14. Different forms of a gene are called alleles. Alleles can be .....dominant..... or recessive. Two copies of a recessive gene are required for the characteristic to be expressed.
15. Gregor Mendel was an Austrian monk who conducted the first quantitative .....genetics..... experiments using garden peas.
16. Punnett squares are a useful tool to predict the proportion of ...genotypes..... and phenotypes in the offspring.

17. Pedigree diagrams are useful in showing how various genes are transferred from one generation to another.

18. DNA fingerprinting is used in forensic investigations and paternity tests.

19. Clones are organisms that are genetically identical